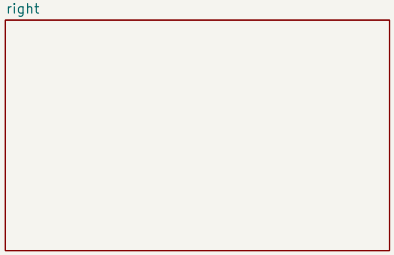
Keygen: one PCB carrying both halves joined by snap-off tabs.
Left half = references 1xx, right half = 2xx. Each half has its own nets (separate GND, VCC).

left



File: left.kicad_sch

right



File: right.kicad_sch

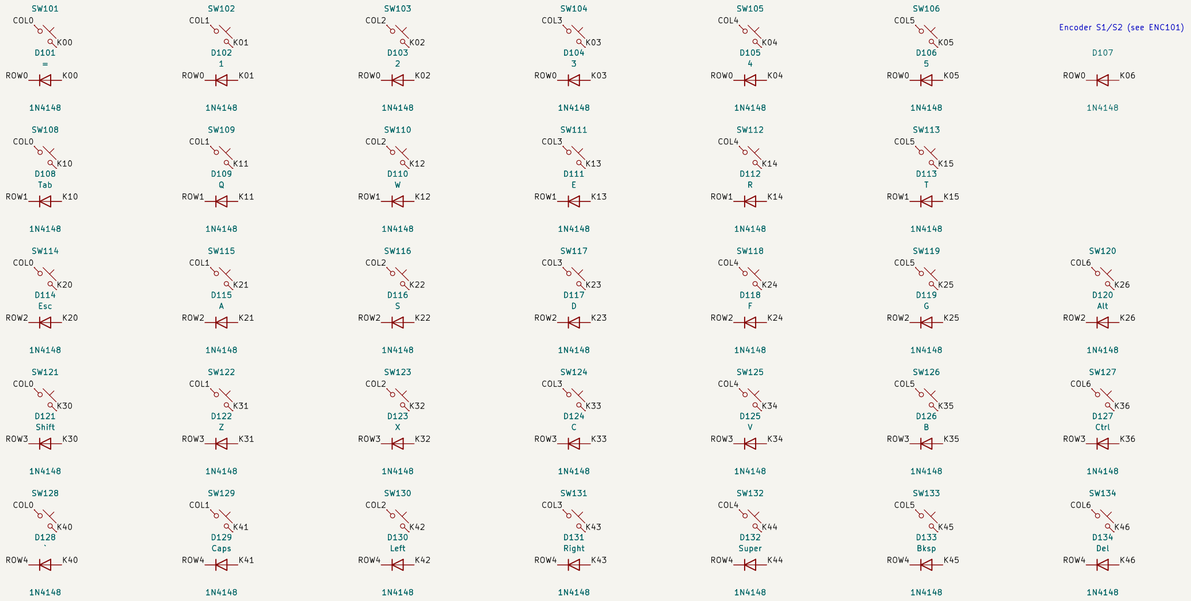
Generated by tools/gen_schematic.py — edit the script, not this file
github.com/milindmadhukar/keygen

Sheet: /
File: keygen.kicad_sch

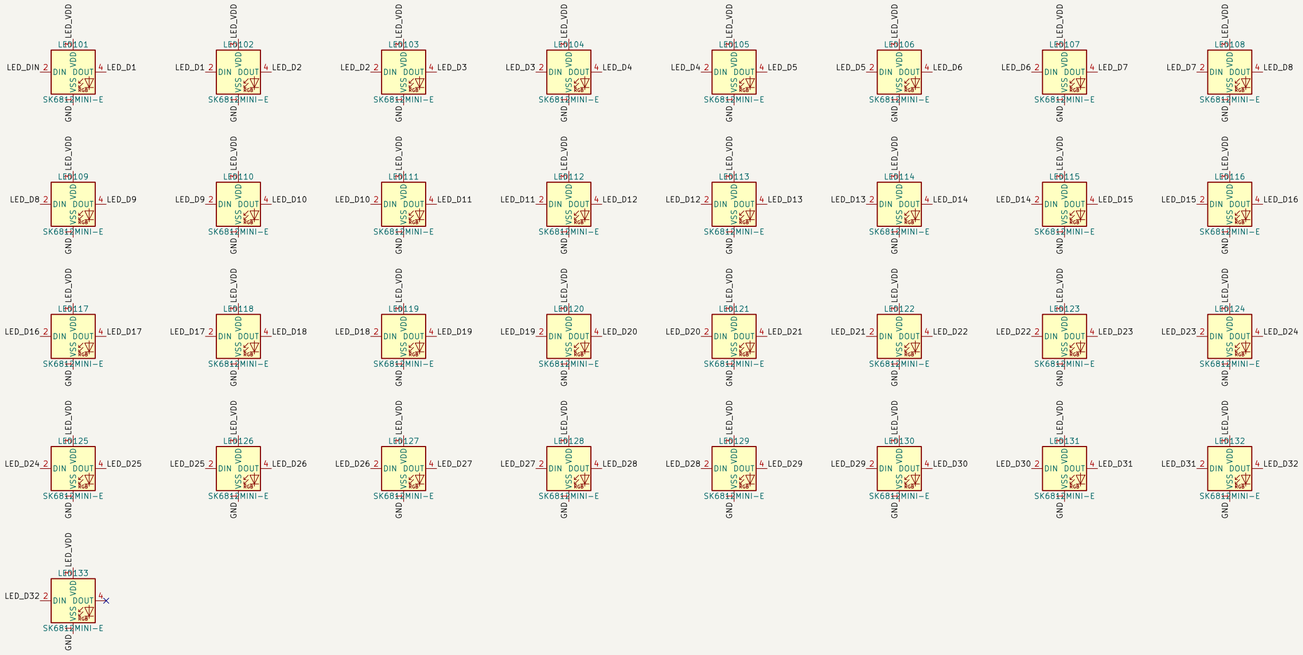
Title: Keygen

Size: A4	Date: 2026-10-20	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 1/3

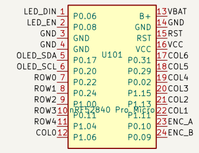
KEY MATRIX (COL2ROW: switch to COLn, diode cathode to ROWn)



PER-KEY LEDs (33) — powered from the battery through the LED power switch (Q2, LED_VDD)

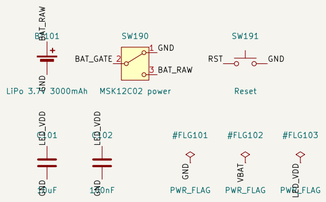


CONTROLLER (mounted on the PCB underside, USB-C to the top edge)



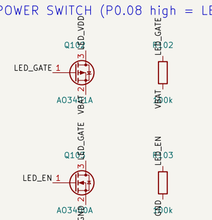
Encoder A/B on P0.10/P0.09 (NFC pins): set nfct-pins-as-gpios.
P0.08: LED power switch. P0.17/P0.20: OLED SDA/SCL.
VCC (pad 16, switched by P0.13): OLED power; firmware holds P0.13 high.

POWER



C1/C2 sit at the LED power switch's output.

LED POWER SWITCH (P0.08 high = LEDs on)

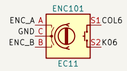


BATTERY SWITCH (P-MOSFET; the slide switch only drives the gate)



Switch 2–1: gate to GND, ON. Switch 2–3: gate to source, off.
R1 holds the gate off between positions. Leave the switch on while charging:
off, USB charges the cell through the body diode (~0.35 V at 500 mA).

ROTARY ENCODER (EC11) — push switch in the matrix at ROW0/COL5



OLED (0.96 in, 128x64 SSD1306, I2C address 0x3C) — check the module's pin order: GND VCC SCL SDA



MOUNTING HOLES (M2)



Generated by tools/gen_schematic.py — edit the script, not this file
github.com/milindmadhukar/keygen

Sheet: /left/		
File: left.kicad_sch		
Title: Keygen — left half		
Size: A2	Date: 2026-10-19	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 2/3

KEY MATRIX (COL2ROW: switch to COLn, diode cathode to ROWn)

SW201 COL0 K00 D201 – ROW0 K00 1N4148
SW202 COL1 K01 D202 0 ROW0 K01 1N4148
SW203 COL2 K02 D203 9 ROW0 K02 1N4148
SW204 COL3 K03 D204 8 ROW0 K03 1N4148
SW205 COL4 K04 D205 7 ROW0 K04 1N4148
SW206 COL5 K05 D206 6 ROW0 K05 1N4148
Encoder S1/S2 (see ENC201)
SW207 D207 ROW0 K06 1N4148

SW208 COL0 K10 D208 \ ROW1 K10 1N4148
SW209 COL1 K11 D209 P ROW1 K11 1N4148
SW210 COL2 K12 D210 O ROW1 K12 1N4148
SW211 COL3 K13 D211 I ROW1 K13 1N4148
SW212 COL4 K14 D212 U ROW1 K14 1N4148
SW213 COL5 K15 D213 Y ROW1 K15 1N4148

SW214 COL0 K20 D214 ' ROW2 K20 1N4148
SW215 COL1 K21 D215 ; ROW2 K21 1N4148
SW216 COL2 K22 D216 L ROW2 K22 1N4148
SW217 COL3 K23 D217 K ROW2 K23 1N4148
SW218 COL4 K24 D218 J ROW2 K24 1N4148
SW219 COL5 K25 D219 H ROW2 K25 1N4148
SW220 COL6 K26 D220 Alt ROW2 K26 1N4148

SW221 COL0 K30 D221 Shift ROW3 K30 1N4148
SW222 COL1 K31 D222 / ROW3 K31 1N4148
SW223 COL2 K32 D223 . ROW3 K32 1N4148
SW224 COL3 K33 D224 . ROW3 K33 1N4148
SW225 COL4 K34 D225 M ROW3 K34 1N4148
SW226 COL5 K35 D226 N ROW3 K35 1N4148
SW227 COL6 K36 D227 Ctrl ROW3 K36 1N4148

SW228 COL0 K40 D228] ROW4 K40 1N4148
SW229 COL1 K41 D229 [ROW4 K41 1N4148
SW230 COL2 K42 D230 Down ROW4 K42 1N4148
SW231 COL3 K43 D231 Up ROW4 K43 1N4148
SW232 COL4 K44 D232 Fn ROW4 K44 1N4148
SW233 COL5 K45 D233 Space ROW4 K45 1N4148
SW234 COL6 K46 D234 Enter ROW4 K46 1N4148

PER-KEY LEDs (33) – powered from the battery through the LED power switch (Q2, LED_VDD)

LED201 LED_DIN LED_D1 LED_VDD GND SK6812MINI-E
LED202 LED_D1 LED_D2 LED_VDD GND SK6812MINI-E
LED203 LED_D2 LED_D3 LED_VDD GND SK6812MINI-E
LED204 LED_D3 LED_D4 LED_VDD GND SK6812MINI-E
LED205 LED_D4 LED_D5 LED_VDD GND SK6812MINI-E
LED206 LED_D5 LED_D6 LED_VDD GND SK6812MINI-E
LED207 LED_D6 LED_D7 LED_VDD GND SK6812MINI-E
LED208 LED_D7 LED_D8 LED_VDD GND SK6812MINI-E
LED209 LED_D8 LED_D9 LED_VDD GND SK6812MINI-E
LED210 LED_D9 LED_D10 LED_VDD GND SK6812MINI-E
LED211 LED_D10 LED_D11 LED_VDD GND SK6812MINI-E
LED212 LED_D11 LED_D12 LED_VDD GND SK6812MINI-E
LED213 LED_D12 LED_D13 LED_VDD GND SK6812MINI-E
LED214 LED_D13 LED_D14 LED_VDD GND SK6812MINI-E
LED215 LED_D14 LED_D15 LED_VDD GND SK6812MINI-E
LED216 LED_D15 LED_D16 LED_VDD GND SK6812MINI-E
LED217 LED_D16 LED_D17 LED_VDD GND SK6812MINI-E
LED218 LED_D17 LED_D18 LED_VDD GND SK6812MINI-E
LED219 LED_D18 LED_D19 LED_VDD GND SK6812MINI-E
LED220 LED_D19 LED_D20 LED_VDD GND SK6812MINI-E
LED221 LED_D20 LED_D21 LED_VDD GND SK6812MINI-E
LED222 LED_D21 LED_D22 LED_VDD GND SK6812MINI-E
LED223 LED_D22 LED_D23 LED_VDD GND SK6812MINI-E
LED224 LED_D23 LED_D24 LED_VDD GND SK6812MINI-E
LED225 LED_D24 LED_D25 LED_VDD GND SK6812MINI-E
LED226 LED_D25 LED_D26 LED_VDD GND SK6812MINI-E
LED227 LED_D26 LED_D27 LED_VDD GND SK6812MINI-E
LED228 LED_D27 LED_D28 LED_VDD GND SK6812MINI-E
LED229 LED_D28 LED_D29 LED_VDD GND SK6812MINI-E
LED230 LED_D29 LED_D30 LED_VDD GND SK6812MINI-E
LED231 LED_D30 LED_D31 LED_VDD GND SK6812MINI-E
LED232 LED_D31 LED_D32 LED_VDD GND SK6812MINI-E
LED233 LED_D32 LED_VDD GND SK6812MINI-E

CONTROLLER (mounted on the PCB underside, USB-C to the top edge)

U201 nice!nano v2 (nRF52840)
LED_DIN 1 P0.06 · LED_EN 2 P0.08 · GND 3 GND · GND 4 GND · OLED_SDA 5 P0.17 · OLED_SCL 6 P0.20 · ROW0 7 P0.22 · ROW1 8 P0.24 · ROW2 9 P1.00 · ROW3 10 P0.11 · ROW4 11 P1.04 · COL0 12 P1.06
B+ 13 VBAT · GND 14 GND · RST 15 RST · VCC 16 VCC · P0.31 17 COL6 · P0.29 18 COL5 · P0.02 19 COL4 · P1.15 20 COL3 · P1.13 21 COL2 · P1.11 22 COL1 · P0.10 23 ENC_A · P0.09 24 ENC_B

Encoder A/B on P0.10/P0.09 (NFC pins): set nfct-pins-as-gpios.
P0.08: LED power switch. P0.17/P0.20: OLED SDA/SCL.
VCC (pad 16, switched by P0.13): OLED power; firmware holds P0.13 high.

POWER

BAT1 BAT_RAW GND — LiPo 3.7V 3000mAh
SW290 MSK12C02 power: GND · BAT_RAW · BAT_GATE 2
SW291 Reset: RST · GND
C201 LED_VDD GND 10uF
C202 LED_VDD GND 150nF
#FLG201 GND PWR_FLAG
#FLG202 VBAT PWR_FLAG
#FLG203 LED_VDD PWR_FLAG

C1/C2 sit at the LED power switch's output.

LED POWER SWITCH (P0.08 high = LEDs on)

Q2 AO3401A: LED_GATE 1 · LED_VDD 2 · VBAT 3
R202 LED_GATE VBAT 10k
Q3 AO3401A: LED_EN 1 · LED_GATE 2 · GND 3
R203 LED_EN GND 10k

BATTERY SWITCH (P-MOSFET; the slide switch only drives the gate)

Q201 AO3401A: BAT_GATE 1 · BAT_RAW 2 · VBAT 3
R201 BAT_GATE BAT_RAW 1M

Switch 2–1: gate to GND, ON. Switch 2–3: gate to source, off.
R1 holds the gate off between positions. Leave the switch on while charging:
off, USB charges the cell through the body diode (~0.35 V at 500 mA).

ROTARY ENCODER (EC11) – push switch in the matrix at ROW0/COL5

ENC201 EC11: ENC_A A · GND C · ENC_B B · S1 COL6 · S2 KO6

OLED (0.96 in, 128x64 SSD1306, I2C address 0x3C) – check the module's pin order: GND VCC SCL SDA

J201 OLED 4pin I2C: GND 1 · VCC 2 · OLED_SCL 3 · OLED_SDA 4

MOUNTING HOLES (M2)

H201 M2 · H202 M2 · H203 M2 · H204 M2 · H205 M2

Generated by tools/gen_schematic.py – edit the script, not this file
github.com/milindmadhukar/keygen
Sheet: /right/
File: right.kicad_sch
Title: Keygen – right half
Size: A2 | Date: 2026-10-19 | Rev: 0.1
KiCad E.D.A. 10.0.7 | Id: 3/3